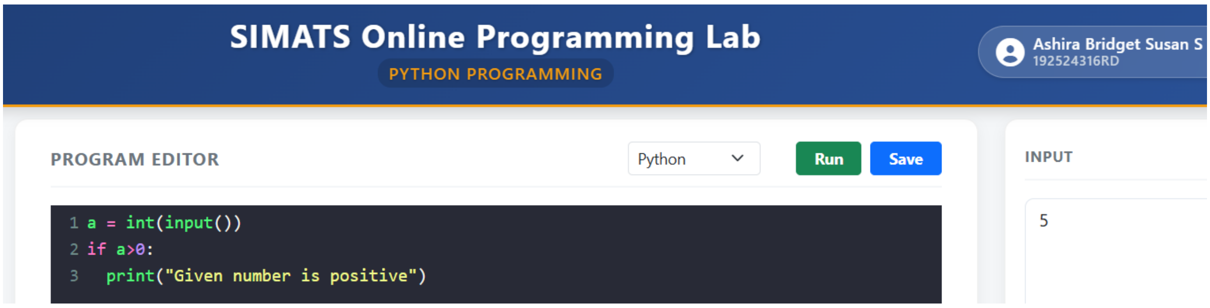
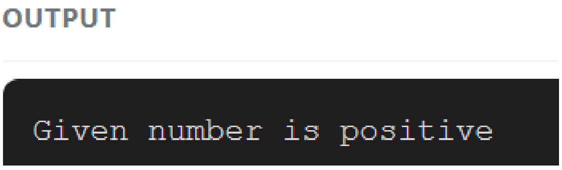
<u>CONTROL STATEMENTS</u>:-

“If” – statements:

1. WAP to check if a given number if positive.





2. WAP to provide flat Rs. 500 if the purchase amount is greater than 2000.

3. WAP to provide bonus mark if the category is sports.

OUTPUT

```
Bonus marks provided
```

4. WAP to check if the number is Zero.

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PROGRAM EDITOR

Python | Run | Save

```python
a = int(input())
if a==0:
  print("Zero")
```

INPUT

```
0
```

OUTPUT

```
Zero
```

“If-Else” – Statement:-

5. WAP to check whether the given years is a leap year or not.

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PROGRAM EDITOR   Python   ✓ Saved   Run   Save

```python
year = int(input())
if (year%4==0 and year%100!=0) or (year%400==0):
  print(year, " is a leap year")
else:
  print(year, " is not a leap year")
```

INPUT

2024

OUTPUT

```
2024  is a leap year
```

6. WAP to check the eligibility for voting.

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PROGRAM EDITOR   Python   Run   Save

```python
age = int(input())
if age>=18:
  print("Eligible for voting")
else:
  print("Not eligible for voting")
```

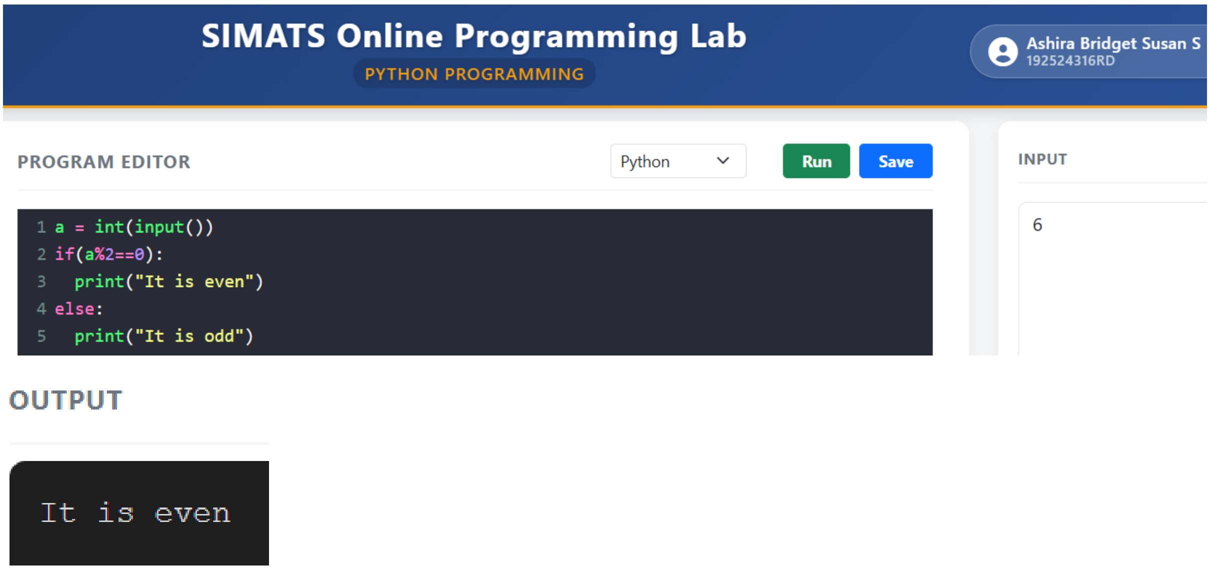
INPUT

27

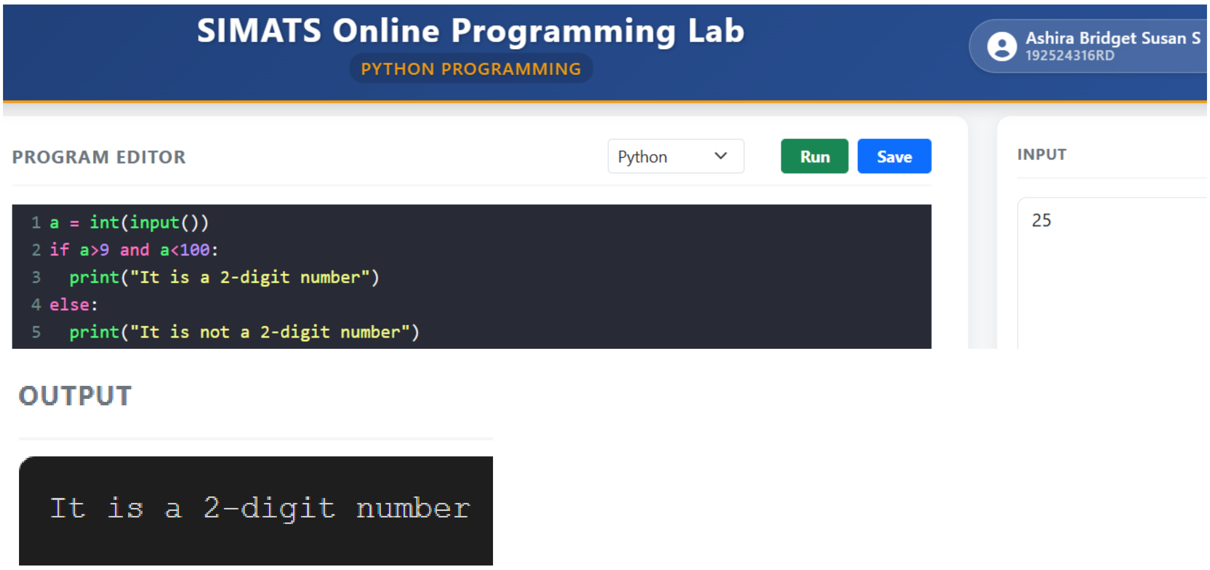
OUTPUT

```
Eligible for voting
```

7. WAP to check if even/odd



8. WAP to check if the given is a 2 digit number or not.



9. WAP to check if a number is divisible by 3 and 5 or not.

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PROGRAM EDITOR   Python   Run   Save

```python
a = int(input())
if a%3==0 and a%5==0:
  print(a, "is divisible by 3 or 5")
else:
  print(a, "is not divisible by 3 or 5")
```

INPUT

15

OUTPUT

```
15 is divisible by 3 or 5
```

10. WAP to check if a given number is divisible by 5 or not, if not, print the remainder.

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PROGRAM EDITOR   Python   Run   Save

```python
a = int(input())
if a%5==0:
  print(a, "is divisible by 5")
else:
  print(a, "is not divisible by 5 and the remainder is ", (a//5))
```

INPUT

6

OUTPUT

```
6 is not divisible by 5 and the remainder is  1
```

11. WAP to see of a number is positive/negative.

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PROGRAM EDITOR

Python Run Save

```python
a = int(input())
if(a>0):
  print("It is positive")
else:
  print("It is negative")
```

INPUT

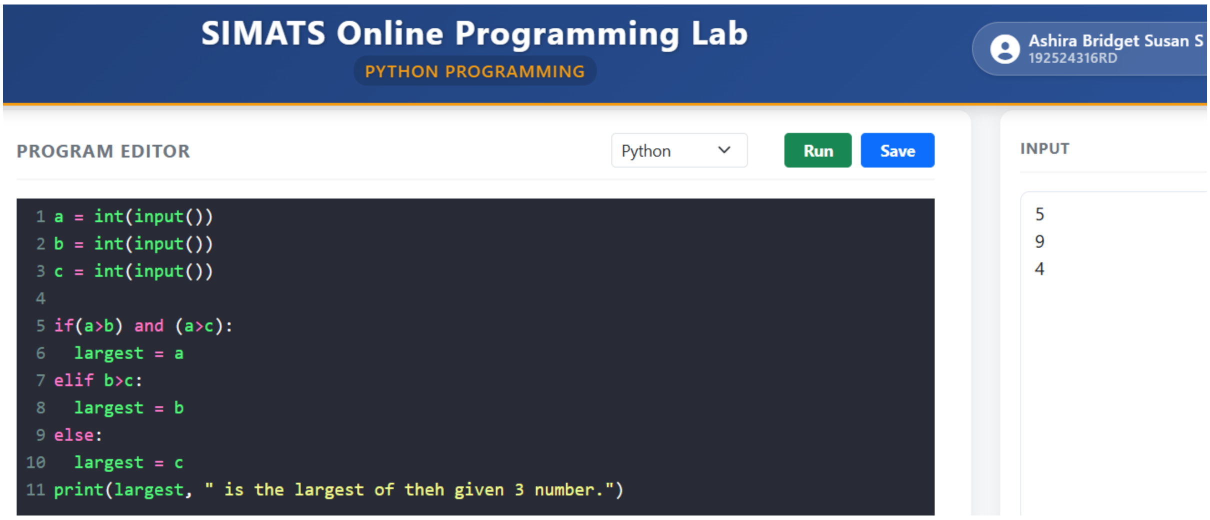
6

OUTPUT

```
It is positive
```

“If-Elif-Else” – statements:-

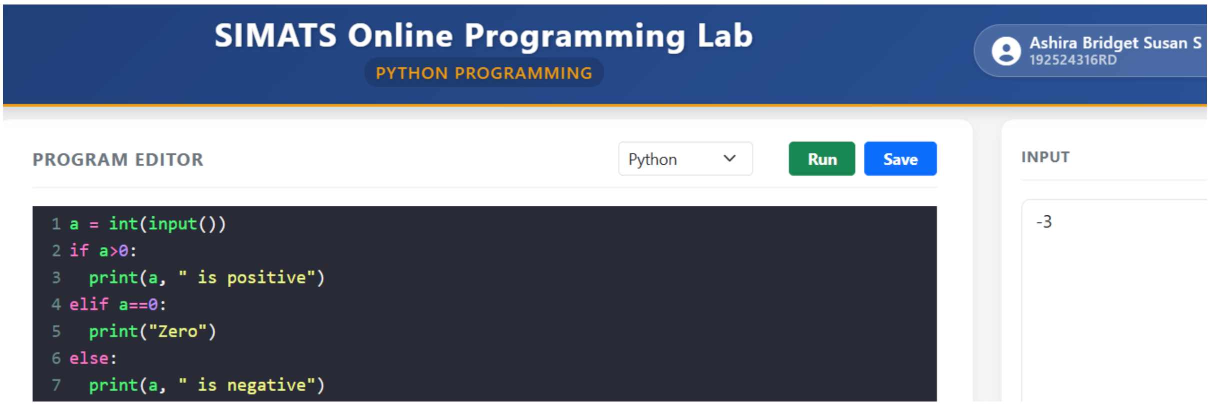
12. WAP to find the largest of given 3 numbers.



OUTPUT

```
9  is the largest of theh given 3 number.
```

13. WAP to check the given number is positive or negative or zero.



OUTPUT

```
-3  is negative
```

14. WAP to check if a number is divisible by 5 and 10 or by 5 or 10.

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PROGRAM EDITOR Python Run Save

```
a = int(input())
if a%5==0 and a%10==0:
  print(a, "is divisible by 5 and 10")
elif a%5==0 or a%10==0:
  print(a, "is divisible by 5 or 10")
else:
  print(a, "is not divisible 5 or 10")
```

INPUT

50

OUTPUT

```
50 is divisible by 5 and 10
```

15. WAP to see if a given number is single/double/above 2/negative no.

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PROGRAM EDITOR Python Run Save

```
a = int(input())
if a>=0 and a<=9:
  print("It is a single digit")
elif a>9 and a<100:
  print("It is a double digit")
elif a>2:
  print("It is above 2")
else:
  print("It is a negative number")
```

INPUT

-9

OUTPUT

```
It is a negative number
```

16. WAP to check the age category.

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PROGRAM EDITOR

Python Run Save

```python
age = int(input())
if age < 12:
  print("You are a child!")
elif age >12 and age<=19:
  print("You are a teenager")
elif age>19 and age<=35:
  print("You are a young adult")
else:
  print("You are a senior citizen")
```

INPUT

66

OUTPUT

```
You are a senior citizen
```

17. WAP for a student Mark System

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PROGRAM EDITOR

Python Run Save

```python
a = int(input())
if a>= 90:
  print("S Grade")
elif a>=80:
  print("A Grade")
elif a>=70:
  print("B Grade")
elif a>=60:
  print("C Grade")
elif a>=50:
  print("D Grade")
else:
  print("F Grade")
```

INPUT

95

OUTPUT

```
S Grade
```

18. WAP for a Traffic Light System

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PYTHON PROGRAMMING

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PROGRAM EDITOR

Python Run Save

```
color = input()
if color == "red":
  print("STOP!")
elif color == "yellow":
  print("SLOW DOWN/GET READY!")
elif color == "green":
  print("GO!")
else:
  print("INVALID COLOR!")
```

INPUT

```
blue
```

OUTPUT

```
INVALID COLOR!
```

19. WAP to find the Roots of Quadratic Equation

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PROGRAM EDITOR

Python Run Save

```
a = float(input())
b = float(input())
c = float(input())

d = (b**2) - (4*a*c)

if d>0:
  r1 = (-b + (d**0.5)) / (2*a)
  r2 = (-b - (d**0.5)) / (2*a)
  print("Two real roots: ", r1, "and", r2)
elif d==0:
  r = -b / (2*a)
  print("One real root:", r)
else:
  print("The roots are imaginary/complex")
```

INPUT

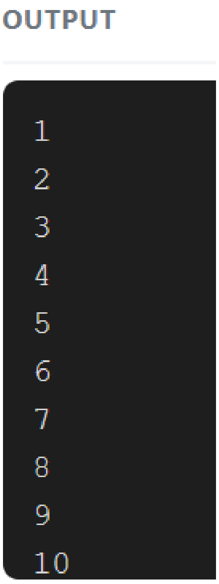
```
1
-5
6
```

OUTPUT

```
Two real roots:  3.0 and 2.0
```

“While” – Loop Statements:-

1. Display 1 to 10



2. Display 10 to 1

OUTPUT

```
10 9 8 7 6 5 4 3 2 1
```

3. Display odd numbers from 1 to 20

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PROGRAM EDITOR
Python
Run
Save

```
num = 1
while(num <=20):
  if(n%2!=0):
    print(num, end = " ")
    num+=1
```

INPUT

5

4. WAPP to find sum of n numbers.

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PROGRAM EDITOR
Python
Run
Save

```
i = 0
sum = 0
while (i <=5):
  sum = sum + i
  i = i + 1
print("Sum of series: ", sum)
```

INPUT

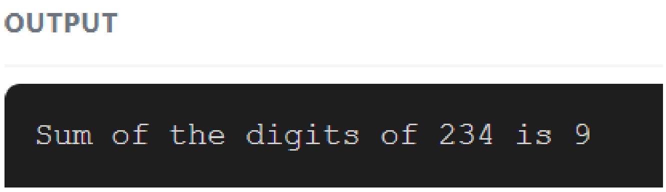
Your INPUT goes here!

OUTPUT

```
Sum of series:  15
```

5. WAPP to find the sum of even numbers.

6. Write a Python Program to find sum of digits of a given number



7. WAPP to print the Fibonacci Series.

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PROGRAM EDITOR

Python Run Save

```python
n = int(input())
a = 0
b = 1
count = 0
print("Fibonacci Series: ")
while(count<n):
  print(a, end = " ")
  c = a + b
  a = b
  b = c
  count += 1
```

INPUT

10

OUTPUT

```
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34
```

8. WAPP to reverse the given number.

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PROGRAM EDITOR

Python Run Save

```python
n = int(input())
rev = 0
while(n>0):
  a=n%10
  rev=rev*10+a
  n=n//10
print(rev)
```

INPUT

12345

OUTPUT

```
54321
```

9. WAPP to find whether the number is armstrong or not.

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PROGRAM EDITOR

Python | Run | Save

```python
n = int(input())
org=n
sum = 0
while(n>0):
  a=n%10
  sum=sum+(a**3)
  n=n//10
print(sum)

if sum==org:
  print("It is an Armstrong Number")
else:
  print("It is not an Armstrong Number")
```

INPUT

153

OUTPUT

```
153
It is an Armstrong Number
```

10. Check number is palindrome or not.

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PROGRAM EDITOR

Python | Run | Save

```python
n = int(input())
org=n
rev = 0
while(n>0):
  a=n%10
  rev=rev*10+a
  n=n//10
print(rev)

if rev==org:
  print("It is a Palindrome")
else:
  print("It is not a Palindrome")
```

INPUT

12521

```
12521
It is a Palindrome
```